

Current and Future Approaches to Pediatric Anxiety Disorder Treatment

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This overview critically appraises the literature on the treatment of pediatric anxiety disorders. The two established treatments for these conditions comprise cognitive-behavioral therapy (CBT) and antidepressant medications. Many youths receiving these treatments fail to achieve remission, which creates a need for new treatments. After summarizing the literature on CBT and currently available medications, the

authors describe research that lays a foundation for improvements in the treatment of pediatric anxiety disorders. This foundation leverages neuroscientific investigations, also described in the overview, which provide insights on mechanisms of successful treatment.

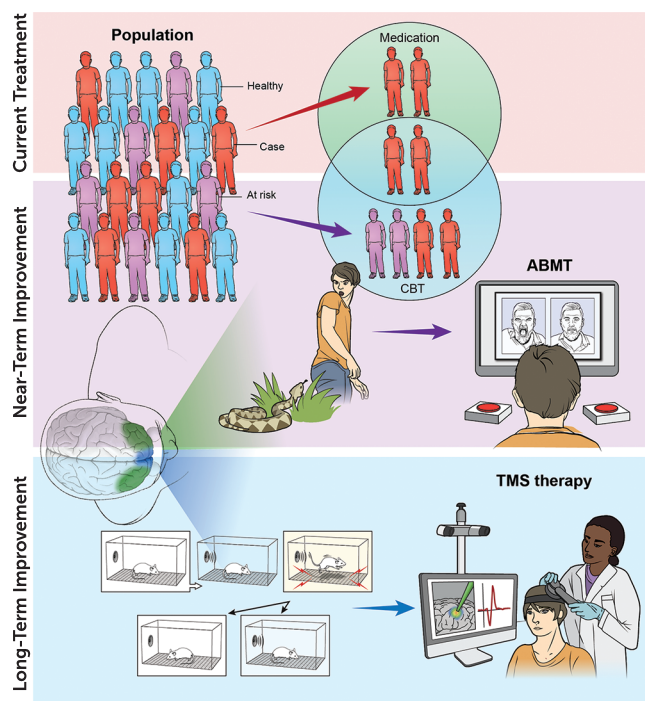
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In this overview, we describe research on three pediatric anxiety disorders: generalized anxiety disorder, separation anxiety disorder, and social anxiety disorder. We consider these disorders as a group, because most treatment studies target them together, given their high prevalence and frequent comorbidity. We omit specific phobia, the fourth common anxiety disorder, which is associated with lower rates of persistence and impairment as well as better long-term outcomes than the three conditions covered here (1). A fifth diagnosis, selective mutism, often occurs comorbidly with social anxiety disorder. Because large clinical trials include children with selective mutism only when they also have social anxiety disorder, data on treatment of social anxiety disorder largely apply to the treatment of selective mutism. Similarly, panic disorder is rare before early adulthood, and large clinical trials in youths include this condition only when it occurs alongside generalized anxiety disorder, separation anxiety disorder, or social anxiety disorder. Hence, data on treatment for these disorders also likely apply to panic disorder.

These three pediatric anxiety disorders are highly prevalent and may cause long-term impairment (2). On the one hand, effective treatments exist for the disorders, and many patients have long-term remissions (1, 3). Longitudinal studies find that one-third to one-half of youths with these anxiety disorders are free of psychopathology when evaluated over the next 10 years (1, 4). This provides evidence of a better prognosis than for most other forms of pediatric psychopathology, including autism spectrum disorders, attention deficit hyperactivity disorder, and major depressive disorder (5). On the other hand, even with adequate treatment, many patients remain symptomatic; other youths relapse after

robust initial responses; and most adults with persistent mood or anxiety disorders previously had a pediatric anxiety disorder. Thus, clinicians confront the duality of a relatively good prognosis with adequate treatment for one set of youths (1, 4) alongside a high risk for chronicity in sizable groups of other youths. Clinicians approaching the treatment of pediatric anxiety disorders can rely on two established treatments: antidepressants and cognitive-behavioral therapy (CBT). In the first two sections of this overview, we summarize research on these established treatments and novel approaches for improving them that may make them more widely applicable. In the third section, we briefly outline a path for finding other innovative treatments that will likely take longer to be widely applied.

The discussion of novel approaches for both relatively immediate and longer-term improvements in treatment leverages a robust literature on the pathophysiology of anxiety disorders. As reviewed elsewhere (6–12), this literature delineates the unique opportunities for therapeutic discovery from the melding of preclinical and clinical research, given strong cross-species conservation in circuitry deployed by mammals when they confront threats. Research on pediatric anxiety disorders finds many signs of dysfunction in this threat-responsive neural circuitry. Signs of such dysfunction manifest at many levels. They include blood-based biomarkers such as cortisol, C-reactive protein, and interleukins that are sensitive to threat exposure (13), neuropsychological and psychophysiological indicators of information processing functions that mammals deploy in the presence of threats (9), and measures of brain structure and function implicated in the mammalian threat response (10, 12). Rather than providing a comprehensive

FIGURE 1. Treatment of and research on anxiety disorders^a

^a Current treatment for individuals diagnosed with an anxiety disorder (red) consists of either medication, CBT, or a combination of both. Near-term improvements in outcomes might follow by increasing availability of these treatments and by offering psychological treatment to people at risk (purple) for anxiety disorders. Current research seeks novel treatments based on their capacity to influence mammalian neural circuitry that is engaged by threats. The figure illustrates two behaviors and associated circuitry. One behavior concerns the capture of attention by a threat, such as the snake in the picture. This behavior engages the lateral prefrontal cortex (shown in green), and may be targeted by cognitive training, as illustrated in the figure. The other behavior appears in the bottom of the figure, where a mouse first learns that a sound predicts a shock before the mouse next extinguishes this learning. Retention of this subsequent extinction learning is represented by the associated sound that appears without the shock. Such learning requires the medial prefrontal cortex (shown in blue). This circuit could be targeted by brain stimulation, as also shown in the figure. ABMT=attention bias modification therapy; CBT=cognitive-behavioral therapy; TMS=transcranial magnetic stimulation.

review here, we narrowly describe research on pathophysiology that provides promising targets for novel treatments. Such research relates modifiable biomarkers to the course of anxiety disorders and describes how attempts to alter these biomarkers for therapeutic purposes might enhance treatment responses in pediatric anxiety disorders.

ESTABLISHED TREATMENTS

The current treatment of anxious youths is mostly conducted in traditional clinical settings, where, after the diagnosis of a disorder is made, treatment consists of one of three modalities: selective serotonin reuptake inhibitors (SSRIs), CBT, or a combination of SSRIs and CBT (Figure 1). Table 1 summarizes relevant meta-analyses published in the past 5

years on these modalities, and this section highlights findings for each treatment.

Pharmacology

Pharmacological treatment represents one of two established therapies for pediatric anxiety disorders. Meta-analyses provide robust evidence of efficacy for virtually all SSRIs and serotonin-norepinephrine reuptake inhibitors (SNRIs) in the treatment of generalized anxiety disorder, social anxiety disorder, and separation anxiety disorder. Nevertheless, only two medications, duloxetine and escitalopram, have received approval from the Food and Drug Administration (FDA) for pediatric anxiety, both for generalized anxiety disorder.

Strong evidence derives both from meta-analyses (summarized in Table 1) and from notable individual randomized controlled trials (RCTs). Perhaps the most influential trial is the Child/Adolescent Anxiety Multimodal Study (CAMS) (14). CAMS included 488 children and adolescents randomly assigned to one of four groups: monotherapy with placebo, sertraline, or CBT, or combined CBT/sertraline therapy. Treatment with sertraline used a flexible dosing design, with a maximal dose of 200 mg/day. For the sertraline monotherapy group, the effect size was 0.45 (95% CI=0.17, 0.74), and the number needed to treat (NNT) was 3.2 (95% CI=3.2, 3.5). These results are comparable with those of other RCTs for SSRIs (15,16) and SNRIs (17). The meta-analytic evidence summarized in Table 1 documents larger effect sizes for SSRIs than for SNRIs (3,18), with no clear superiority for any drug within a class. Across many pharmacological studies, worse treatment response is related to higher baseline symptom severity (19) and a diagnosis of social anxiety disorder (20). Nevertheless, despite these notable beneficial effects, less than half of patients receiving antidepressants have long-term remissions. In the CAMS trial, only about 20% of patients receiving any treatment displayed consistently low levels of anxiety over a 4-year follow-up period, with no differences among treatments, and in the group receiving sertraline, only 40%–52% of patients were in remission at any of the four yearly follow-up assessments (4). Thus, while medications reduce symptoms in the short term, a need exists for new therapies that might produce long-term remission.

The use of pharmacological treatment carries the risk of adverse effects. Behavioral activation is one of the most common notable effects, and it appears to be more common with SSRIs than with SNRIs. It typically occurs at rates below 15%. Antidepressant labeling includes a black box warning of an increased risk of suicide among individuals under age 25. This warning stems from a meta-analysis of industry-sponsored clinical trials that found a moderate effect size for suicidality (21). The reported risk in the FDA analysis was higher for treatment with paroxetine or venlafaxine, as compared to other antidepressants. Other data suggest that paroxetine use may be associated with a relatively high risk for thoughts about suicide (18). Despite

TABLE 1. Summary of meta-analyses of child and adolescent anxiety treatment published 2019–2023^a

Meta-Analysis	Study	Studies (N)	Subjects (N)	Age (years)	Comparison	Effect Size	95% CI
CBT	James et al. (28)	87	5,964	2–19	Waiting list/no treatment, TAU, attention control, alternative treatment, and medication	OR=5.45	3.90, 7.60
	Baker et al. (29)	16	766	11–18	Active versus passive control	SMD=0.454 ^b	0.22, 0.69
CBT, group therapy	Sigurvinsdóttir et al. (101)	81	3,386	<18	Waiting list	OR=8.96 ^c	4.03, 19.90
SSRI	Dobson et al. (18)	20	2,623	5–17	Placebo	SMD=5.2	2.8, 8.8 ^d
SNRI	Dobson et al. (18)	20	2,623	5–17	Placebo	SMD=2.5	–0.1, 5.1
Placebo response	Motta et al. (102)	135	12,583	N/A	SSRI/SNRI	SMD=–1.11	–1.22, –1.00

^a CBT=cognitive-behavioral therapy; OR=odds ratio; SMD=standard mean difference; SNRI=serotonin-norepinephrine reuptake inhibitor; SSRI=selective serotonin reuptake inhibitor; TAU=treatment as usual.

^b Reduction of symptoms.

^c No significant differences compared with individual CBT.

^d Credible interval.

potential differences among particular medications, clinicians should still carefully monitor suicidality in patients receiving any SSRI or SNRI.

Other classes of medication are commonly used to treat anxiety in adults, and some of these have been investigated in children and adolescents. However, these medications should be considered second- or third-line options. Tricyclic antidepressants appear to be less effective than SSRIs while having more concerning side effects (18). Moreover, they are more lethal in overdose than SSRIs and SNRIs, making them third-line options. Benzodiazepines are prescribed because of their rapid onset of action and a wealth of data in adults, although only three small RCTs have been conducted in youths (22). These studies included small samples, with only 30 patients treated with alprazolam in the largest RCT (23). Additionally, meta-analyses provide no evidence of efficacy, showing a small effect size and a wide confidence interval (odds ratio=1.4, 95% CI=0.3, 6.1) (18). Benzodiazepines also include problematic side effects in youths, including a risk for disinhibition and harmful impacts on cognition. Nevertheless, given the underpowered nature of existing studies and notable effects in adults, it is reasonable to consider treatment with benzodiazepines, particularly in acute scenarios where short-term use is needed.

Finally, buspirone and guanfacine represent two other medications studied in RCTs with more than 50 patients. For buspirone, one RCT with 227 patients and another with 341 patients examined efficacy (24), whereas for guanfacine, one RCT with 83 patients examined efficacy (25). Across the three trials, primary efficacy analyses show no advantage for either medication. Nevertheless, both medications are well tolerated in youths and are widely used for indications other than pediatric anxiety disorders. Moreover, secondary analyses provide some suggestive evidence of efficacy for both medications. Hence, both medications are reasonable second-line options for patients who fail to respond to multiple SSRIs or SNRIs.

Cognitive-Behavioral Therapy

CBT is an established first-line treatment for pediatric anxiety disorders, with effect sizes comparable to those for antidepressants (26). CBT for anxiety disorders usually targets avoidance through gradual exposure to threatening stimuli combined with psychoeducation and the teaching of coping strategies. Many RCTs examine the efficacy of manual-based CBT, usually delivered across 12–20 sessions in various formats, such as group or parent-focused CBT (27). The effect size for CBT alone in the CAMS trial was 0.31 (95% CI=0.02, 0.59) (14), while meta-analytic evidence finds larger effect sizes favoring CBT over a waiting list condition or treatment as usual (Table 1). Even with moderate effect sizes for CBT, one-half (28) to two-thirds (29) of patients will not achieve full remission. As in pharmacological treatment, higher baseline symptom levels predict worse treatment response, and parental psychopathology may also predict outcome in some settings (30). Notably, CBT involves lower risk for adverse effects than medication, with no serious adverse effects reported in most RCTs. As a result, it is universally viewed as at least one of two possible first-line therapies.

CBT delivered to different age and diagnostic groups involves many similarities. The treatment often begins with the creation of a fear hierarchy, where the patient lists mildly threatening scenarios on the bottom and very frightening scenarios on the top. Next, the therapist teaches the patient coping strategies and other approaches that help the patient reduce their distress when exposed to feared situations. Finally, patients undergo guided exposures across the typical 2- to 3-month course of therapy, where each session involves confronting scenarios on successively higher hierarchy levels. Beyond such similarities, notable differences exist in approaches to CBT among patients with distinct clinical presentations. With separation anxiety disorder, therapists often must work closely with families to craft experiences that allow graded separations from parents, whereas in social

anxiety disorder, therapists might venture out of the office with patients to assist them in performing frightening social tasks, such as purchasing a snack in a crowded store. Unlike separation anxiety and social anxiety disorders, with generalized anxiety disorder, patients often fear abstract scenarios, which may require the therapist to creatively devise effective exposure sessions. For example, in patients who fear failure, therapists might work with patients to purposely make minor errors in a homework assignment or perform an after-school activity below their level of proficiency. Thus, these differences in treatment reflect unique clinical scenarios.

Combination Treatment

More research is needed comparing SSRI and CBT monotherapy to each other and to combined SSRI/CBT therapy. On their own, effect sizes over control therapies across distinct studies appear comparable for SSRIs and CBT. However, such comparisons can be misleading, since medication and psychotherapy trials can recruit different types of patients and rely on different control therapies. Given comparable efficacy and more concern with adverse effects with SSRIs, some European guidelines (31) consider CBT to be the only first-line therapy, unlike some U.S. guidelines, which consider the two treatments to be equivalent first-line options (32). Guidelines uniformly recommend combined treatment in patients who fail to respond to monotherapy.

Only two notable RCTs of combined treatment have been conducted. First, CAMS found the combination of CBT and medication to be superior to either modality alone (14). However, patients who received combination treatment were told that they were randomized to active medication, and no patient groups received CBT with placebo augmentation. Only patients randomized to receive either sertraline or placebo monotherapy received blinded therapy. Hence, combined treatment's superiority could reflect expectancy as opposed to particular effects of SSRIs. Second, consistent with this possibility, another study compared the combination of CBT and SSRI augmentation to CBT and placebo augmentation, in an attempt to equilibrate between-condition expectancy. Unlike in CAMS, the outcome for youths receiving CBT with an SSRI was no better than for those receiving CBT with placebo (26).

The stronger effects observed in CAMS could reflect design issues or differences in sample size, and more research that controls for expectancy with combination treatment is needed. Nevertheless, the effect size in the more recent study equating expectancy was small, suggesting the need for very large sample sizes in future studies (26). Regardless, when expectancies of improvement are equalized, data suggest that SSRI augmentation to CBT may provide no benefit over CBT alone. Hence, many clinicians begin treatment with CBT and augment the treatment with medication in cases where there is no adequate clinical response after 8–12 weeks.

NEAR-TERM IMPROVEMENTS IN TREATMENT OUTCOMES

Figure 1 highlights current treatments and approaches for improving them, informed by the data summarized in Table 2. In this section, we discuss three promising avenues for generating near-term improvements in the treatment of anxiety disorders. One avenue extends the reach of available treatments; the second applies precision medicine approaches; and the third leverages research on mechanisms to be targeted by novel therapies.

Extending Available Treatments

Several approaches exist for maximizing the impact of established treatments. These include techniques for leveraging technology and for reducing the burden faced by primary care providers and educators. As discussed below, another approach focuses on prevention to interrupt, early in the course of a developmental cascade, processes that can generate chronic emotional problems.

Technology provides unique opportunities for addressing the needs of youths with anxiety disorders. Considerable research suggests that technology can increase the availability of CBT and other forms of therapy. Meta-analyses of RCTs of Internet-delivered therapy are summarized in Table 2; the table considers treatment of various conditions given the paucity of RCTs focused specifically on pediatric anxiety disorders. The studies summarized in the table generate two conclusions (33). First, comparable levels of efficacy might be achieved by using technology to deliver CBT in patients' homes, as compared to the clinic. Second, some level of regular therapist contact appears necessary to achieve such efficacy.

If results of future studies confirm existing research, the standard of care may one day involve having patients conduct some facets of CBT while being directly monitored by a therapist and conduct other facets on their own with the aid of technology. Extending this approach from research settings to the clinic could rapidly increase therapist availability, if other limitations in existing work could be addressed. For example, the control condition in studies to date has often been treatment as usual or a waiting list condition. Although data in adults indicate that Internet-delivered CBT may be as effective as face-to-face CBT (34), more large, well-designed studies are needed in children and adolescents that directly compare CBT delivered via telehealth to face-to-face therapy.

Beyond increasing availability of established treatments, technology provides other opportunities for improvement. Studies in adults suggest that therapeutic processes can be monitored using technology in ways that identify barriers to successful CBT outcomes (35). Studies in youths find that better clinical outcomes occur following specific types of therapeutic encounters, such as exposure sessions involving sustained threat confrontation (36). Future research might integratively extend these two observations by using technology

TABLE 2. Novel treatments and approaches to the treatment of anxiety: results from meta-analyses published 2019–2023^a

Meta-Analysis	Study	Studies (N)	Subjects (N)	Age (years)	Effect Size	95% CI
Improving treatment access	Wakefield et al. (103)	47	636,734	>18	Cohen's $d=0.88$	0.79, 0.97
ABMT	Fodor et al. (71)	65	3,879	>18	SMD = -0.55^b	$-0.91, -0.19$
School based	Hugh-Jones et al. (104)	18	2,076	Children and adolescents	Hedges' $g = -0.28^b$	$-0.50, -0.05$
Telehealth	Gee et al. (105)	45	1,075 ^c	10–19	SMD = -0.49^b	$-0.79, -0.19$
	Venturo-Conerly et al. (106)	37	2,054 ^d	Mean = 9.38 (SD = 4.19)	Hedges' $g = 0.62$	0.16, 1.07
Computer-based CBT	Grist et al. (107)	34	3,113	6–18	Hedges' $g = 0.66$	0.42, 0.90
Mindfulness-based stress reduction	Zhou et al. (108)	14	1,489	12–25	SMD = -0.14	$-0.24, -0.04$
Parental CBT	Yin et al. (27)	6	407	Children and adolescents	SMD = -0.72^b	$-1.41, -0.03$
CBT in low- and middle-income countries	Venturo-Conerly et al. (109)	34	4,176	Mean = 13.38 (SD = 2.66)	Hedges' $g = 0.76^e$	$-0.13, 1.65$

^a ABMT = attention bias modification therapy; CBT = cognitive-behavioral therapy; SMD = standard mean difference.

^b Outcome was the mean change, reduction of anxiety symptoms.

^c Subset of anxious children.

^d Total N not reported in the main manuscript; derived from the report's supplemental Table 1.

^e Effect size of anxiety (included studies in posttraumatic stress disorder and obsessive-compulsive disorder) as moderator of treatment response.

to monitor therapeutic processes related to clinical outcomes in pediatric anxiety disorders. With such technology, patients, families, and clinicians might monitor the engagement of processes that predict remission. Finally, mobile technology provides other opportunities through real-time, multi-day monitoring of multiple parameters. In later sections of this review, we describe therapeutic research that targets mechanisms. Wearable devices may facilitate such work by providing abundant data on physiology and sleep patterns, as well as avoidance and other facets of physical activity that may reflect a treatment's impact on a targeted mechanism.

Independent of technology, another opportunity for advancing treatment outcomes leverages patients' regular contact with pediatricians and educators. Considerable research confirms pediatricians' ability to identify anxiety disorders, which supports recent recommendations for in-office anxiety screening during routine well-child visits (37). Enhancing the capacity for in-office treatment delivery could improve outcomes. Among adults, CBT delivered in primary care settings, relative to a control intervention, produces greater reductions in anxiety symptoms that persists for at least 18 months (9, 38). Some research begins to extend this approach to pediatric anxiety disorders (37, 39, 40). However, large-scale RCTs are needed to test whether the embedding of services within primary care settings produces long-term reductions in pediatric anxiety, as it does in adult anxiety disorders. Educators, too, can identify pediatric anxiety disorders, and RCTs have already documented the efficacy of CBT delivered in school settings (41, 42). While more work is needed to confirm efficacy comparable to traditional forms of CBT (43), offering CBT in schools might help participants access treatment while lowering the associated cost and burden. School-based treatment

also could represent one component of stepped care, where youths who fail to respond or who have problems with greater severity are referred to specialized services (40).

A final approach for extending the impact of available treatments utilizes prevention. Pediatric anxiety disorders represent an intermediate step on a developmental cascade that involves early-life risks related to temperament and late-stage consequences that extend beyond anxiety into other emotional realms (44). For example, pediatric anxiety disorders strongly predict risk for mood disorders, which are more likely to persist and are more difficult to treat than pediatric anxiety disorders. Hence, treatments delivered early in the course of this developmental cascade may be more effective than those delivered at later points. From this perspective, enhancing treatment for pediatric anxiety disorders can be viewed as a form of prevention for mood disorders. For universal and selective forms of prevention, this logic supports other approaches for bolstering resilience and assisting high-risk individuals.

Universal prevention targets population-level risk. To approach this goal, some studies deliver brief psychological interventions in schools or through technology by adapting features implemented in CBT and related therapies. Recent RCTs suggest that effect sizes are not large for these interventions, but a small beneficial effect applied to a large population could be important. As such, the scalable nature of these treatments may support broad delivery in ways that produce clinical impact (45–47). Other universal prevention research suggests the importance of targeting poverty, deprivation, and social stressors that accompany these community-level risks in populations of children where anxiety disorders are particularly common (48). Nevertheless, while programs of cash transfer or universal minimal income appear to benefit many aspects of mental health,

weaker or less consistent effects exist for anxiety than for symptoms of behavior disorders and problems with substance use or school failure (2, 49–51).

Figure 1 depicts facets of selective prevention focused on youths at risk for anxiety. In terms of factors amenable to a selective prevention approach, anxiety disorders have one clear targetable risk marker, namely, behavioral inhibition. Behavioral inhibition manifests in the first few years of life as a tendency to react with hesitancy when confronting novel stimuli, scenarios, or people. Behavioral inhibition strongly predicts risk for multiple anxiety disorders, and associations with social anxiety disorder appear particularly robust and well replicated (44, 52). Early work suggests that children with behavioral inhibition benefit from targeted interventions that reduce or prevent the worsening of anxiety and other problems (53).

Precision Medicine

The availability of CBT, antidepressants, and their combination creates situations well-suited for precision medicine approaches. Practitioners rely on impressionistic information or consensus guidelines that vary across countries to choose from among these three options. Precision medicine approaches might generate useful data-driven alternatives to guide, in a more uniform manner, practitioners around the world who face similar choices about treatment options (54).

Longitudinal research and investigations of response moderators in RCTs provide some guidance about predictors of outcome in pediatric anxiety disorders. In multiple studies, high levels of severity, comorbidity, temperamental factors, and social anxiety symptoms all predict relatively poor outcomes (1). These and other factors identified through meta-analysis could be combined through risk calculators to create precision medicine tools, as has been done for other mental illnesses (55). Moreover, emerging strategies of non-supervised machine learning can inform these efforts by leveraging research to cluster subjects according to high-dimensional data. Cluster membership can then inform creation of risk calculators that predict treatment response to CBT or medications. Particularly useful tools would go beyond prediction of overall outcomes by differentially predicting response to CBT, antidepressants, or their combination. However, this will require additional work, since available data from large RCTs fail to identify factors that predict differential responses (56).

Multivariate analysis of brain imaging data also may inform the creation of risk calculators. This approach could leverage resting-state functional MRI (rs-fMRI) to relate brain function to treatment response. One recent study (57) utilized rs-fMRI in adults with major depressive disorder who were randomized to receive either CBT or antidepressant therapy. The study found that distinct rs-fMRI patterns predicted response to these two therapies. Attempts to extend this approach to anxiety must contend with inconsistent findings for rs-fMRI research in patients with anxiety disorders (58). However, multivariate analysis

does relate rs-fMRI to symptoms of pediatric anxiety disorders with some consistency, which sets the stage for future development (59, 60). Of note, these methods require large samples to minimize the risk of overfitting. Existing reports, such as from the Adolescent Brain Cognitive Development (ABCD) study (61), show rs-fMRI to be sufficiently scalable to support this work. Moreover, mega-analyses can pool individuals from multiple samples to generate data sets with greater numbers of patients than in single cohorts, even ones as large as the ABCD study (62–64). Findings from studies adopting these approaches suggest that rs-fMRI, as compared to structural MRI data, may provide more accurate targets for precision medicine. These approaches suggest that rs-fMRI might be used to identify children and adolescents with anxiety disorders most likely to benefit from SSRIs, CBT, or their combination. When embedded in a large RCT comparing these treatment arms, future imaging studies might convincingly relate patterns of functional connectivity to treatment outcomes. If outcomes relate to particular treatment choices, as in preliminary work on major depressive disorder (57), this could reveal how clustering according to certain models produces results superior to treatment as usual.

Beyond rs-fMRI, another useful approach exploits similarities between CBT and extinction, a process where neural circuitry has been mapped in many species (12). The lower left section of Figure 1 depicts an extinction experiment conducted with a rodent. A first experimental phase involves exposure to a threat, a sound paired with a shock. Extinction occurs in the next experimental phase, when the animal learns to treat the sound as neutral, after it no longer predicts the shock. Processes engaged during exposure therapy in CBT resemble processes that allow rodents to reduce defensive response to an extinguished threat (65).

Research extending experiments on extinction to humans incorporates psychophysiology, electrophysiology, and fMRI markers (66). This work links pediatric anxiety disorders to heightened responding to neutral stimuli misperceived as threatening (12, 40). Recent research on extinction in large pediatric samples confirms findings from smaller samples. For example, two studies (67, 68) report positive relationships between levels of physiologic responding to extinguished threats and levels of pediatric anxiety disorder symptoms, and a third study (69) uses fMRI to show that extinguished threats evoke stronger amygdala–ventromedial prefrontal cortex (vmPFC) connectivity in youths with anxiety disorders than in healthy youths. This latter study also reports age differences in amygdala–vmPFC connectivity that resemble age differences reported in other research on extinction, in both rodents and humans (12). Finally, few studies of extinction have examined relations with treatment (40). As in rs-fMRI research, large-scale RCTs are needed that incorporate measures of extinction as potential predictors of treatment response. Such a focus on extinction also informs strategies for creating novel therapies, discussed in the next section.

Targeting Mechanisms

Research on mechanisms contributing to pediatric anxiety disorders provides insights on a multi-phased approach involving scenarios depicted in Figure 1. This approach begins in cross-sectional studies relating pediatric anxiety symptoms to correlates of mechanisms presumed to initiate or sustain anxiety disorders. This is followed by longitudinal research linking future risk for anxiety to such mechanistic correlates, which then also can be targeted in open trials that alter these correlates and examine changes in symptoms. Finally, RCTs evaluate the role of such mechanisms in treatment. The availability of antidepressants and CBT, two effective therapies, informs this approach. Given children's vulnerable state, novel treatments must either possess advantages over these therapies, be associated with a relatively low risk of adverse effects, or be prioritized for use in treatment-refractory patients.

A recent review (11) discussed the need to balance the availability of effective treatments with the need to find new ones for pediatric anxiety disorders. As an example, it highlights the low risk of adverse effects as an important feature of cognitive bias modification, where the approach utilizes attention bias modification therapy (ABMT) to target threat-related attention bias. This type of attention bias occurs when threats, as compared to neutral stimuli, produce greater disruptions in attention. Children expressing high risk for anxiety or ongoing anxiety disorders manifest greater threat-related attention bias than healthy children (11, 40). Various schemes exist for reducing this bias, but the most consistent results have been produced with ABMT that trains children to shift their attention away from threat. While ABMT exerts replicable, statistically significant effects on anxiety, meta-analyses of stand-alone ABMT in both adults and children find modest effect sizes, typically with d values below 0.60 (70, 71). In addition, modest evidence of efficacy exists across three studies that added ABMT to CBT for pediatric anxiety disorders, here with small to medium effect sizes (6, 72, 73). Thus, evidence of efficacy exists, but increasing the magnitude of ABMT efficacy will be important.

We describe research on brain stimulation more fully in the next section, but brain stimulation is mentioned here because preliminary evidence suggests that it could increase the effect size for ABMT. Imaging research implicates dysfunction in the dorsolateral prefrontal cortex (dlPFC) in threat-related attention bias that occurs in pediatric anxiety disorders, as highlighted in the middle section of Figure 1 (74). By hindering patients' ability to maintain neural representations of their goals, dlPFC dysfunction may enable attention bias by allowing mild threats to inappropriately capture attention (75). Data suggest that CBT and ABMT partially alleviate dlPFC dysfunction in pediatric anxiety disorders (74). This therapeutic effect could be enhanced through direct dlPFC stimulation designed to further correct dlPFC dysfunction associated with pediatric anxiety disorders. Consistent with this possibility, a

study in adults (75) shows that active relative to sham dlPFC stimulation during ABMT produces greater changes in attention.

While examined less extensively than threat-related attention bias, other forms of cognitive dysfunction also correlate with pediatric anxiety symptoms in ways that provide additional targets for novel therapies. For example, children with anxiety disorders manifest elevated activity in the ventrolateral prefrontal cortex and associated interconnected areas that comprise the brain's so-called ventral attention network. This anxiety-related enhancement in brain activity occurs when generic, non-emotional, salient stimuli involuntarily capture attention (11). Thus, pediatric anxiety disorders involve a hypersensitivity to distractors on both neural and behavioral grounds that could be reduced through cognitive training designed to alleviate this hypersensitivity. Similarly, Craske and colleagues (76) suggest that deficient inhibitory control, which occurs in pediatric anxiety disorders (77, 78), impedes CBT response by hindering the extinction of conditioned fear. Preliminary work in adults suggests that cognitive training regimens capable of enhancing inhibitory control could improve response to CBT (79).

Research in adults uses many approaches to target mechanisms and improve treatment responses. This research informs studies of pediatric anxiety disorders, since more work in adults than in youths uses pharmacotherapy to target mechanisms. One recent review (80) describes compounds used in this manner, including oxytocin, cortisol, and other modulators of hypothalamic-pituitary-adrenal axis function, as well as D -cycloserine and other medications that modulate glutamate. Like the targeting of inhibitory control deficits with cognitive training, many such compounds target the circuitry supporting extinction of conditioned fear. As discussed above, imaging studies implicate this circuitry in pediatric anxiety disorders, suggesting the relevance of studies in adults for youths. Unfortunately, limited research in youths has employed pharmacotherapy to target extinction, possibly owing to concerns about adverse effects. The best-developed approach uses D -cycloserine to augment CBT in obsessive-compulsive disorder (OCD), a condition related to pediatric anxiety disorders. However, results from an RCT in 142 pediatric patients with OCD suggest that the medication fails to augment response to CBT (81).

Examination of sleep provides a final example where research on mechanisms informs approaches to novel treatment in pediatric anxiety disorders. Problems with sleep correlate with ongoing and future risk for pediatric anxiety disorders (82). At a mechanistic level, such problems may disrupt memory consolidation in ways that hinder extinction of fear. Evidence (83) suggests that novel techniques for consolidating memory during sleep could potentially be adapted to facilitate response to CBT through consolidation of extinction. Similarly, other work has generated a model (82) that links disrupted sleep to pediatric anxiety disorders through effects on functioning of the brain's default mode network. Novel treatments (84) apply this model through

sleep enhancement programs targeting multiple processes. These include distressing thoughts and associated rumination at bedtime in addition to alterations in motivation, deficient savoring of positive experiences, and problematic sleep hygiene. These programs attempt to increase response to CBT in pediatric anxiety disorders by targeting sleep's associated disruptions in brain function.

TREATMENT IMPROVEMENTS INTO THE FUTURE

The above-noted studies of attention bias and extinction illustrate how research on mechanisms provides a useful path for therapeutic discovery involving techniques with low risk of adverse effects. Of note, cognitive training influences brain function in less direct ways than other techniques, such as medications and some forms of brain stimulation, which carry greater risks for adverse effects. This provides avenues for synergistic work that combines research on cognitive training, a technique with low risk of adverse effects, with research on other therapies carrying greater risk for adverse effects. By providing complementary perspectives on mechanisms, such synergistic work could generate saltatory long-term improvements in treatment.

While minimal research has examined brain stimulation in any pediatric mental disorder, let alone pediatric anxiety disorders, research on ABMT summarized in the previous section illustrates the promise of melding studies of cognitive training and brain stimulation. Accordingly, this final section highlights some facets of brain stimulation. Some techniques, such as transcranial direct current stimulation (tDCS), carry relatively few risks and hold some promise, where stimulation of the prefrontal cortex might influence extinction in ways that could facilitate response to CBT. However, one study in healthy adults found unexpected effects of tDCS on extinction: stimulation parameters that were expected to facilitate extinction of conditioned fear generated the opposite effect (85, 86). Unexpected results could reflect a less clear understanding of tDCS's neural effects than for another promising treatment, transcranial magnetic stimulation (TMS).

TMS carries some risk of inducing seizures. While no TMS data exist in pediatric anxiety disorders, overall seizure risk with TMS in people of all ages is well below 1% (87). Work relevant to pediatric anxiety disorders includes an RCT of TMS in 103 pediatric patients with major depressive disorder, where no seizures occurred among the 48 patients who received as many as 30 treatment sessions (88). As described in the section above for ABMT, studies of TMS and cognitive training could inform each other, where TMS could be appropriately delivered to children with refractory disorders. Of note, such work could progress slowly, given findings among adults, where this approach has yet to generate the expected therapeutic benefits for anxiety, beyond successes in the treatment of OCD. For example, reminiscent of the above-noted unexpected tDCS effects on extinction (86), results from a study in 125 adults with posttraumatic stress disorder suggest that TMS stimulation intended to

facilitate extinction could evoke the opposite effect, slowing CBT-related extinction (89).

Increasing discussion between basic and clinical researchers is important. Available work links theories of anxiety disorder pathogenesis to neuroscience research on mammalian defensive behavior (7, 8, 10, 44, 90–95). Moreover, this work specifically highlights the relevance of research in nonhuman primates, given the evolutionary relatedness of nonhuman primates to humans, the prolonged development of the primate brain, and the well-developed prefrontal cortex of primates. This work supports neurodevelopmental theories of anxiety, identifying neural circuits and altered gene expression within the amygdala and prefrontal cortex that appear to be related to risk for anxiety disorders (10, 91–93). As with research on prevention, this work suggests that more durable therapeutic effects may result from earlier treatment, and it also delineates many pathways for clinical research.

Cross-species similarities allow neuroscience to inform a long-term agenda for novel treatment discovery in pediatric anxiety disorders. As reviewed elsewhere (80, 96–98), new treatments could involve novel pharmacological agents, including small molecules capable of targeting circuits or mechanisms related to defensive behavior, compounds that enhance neural plasticity, and medications showing some initial success in research on adult mood and anxiety disorders. However, two factors suggest that brain stimulation provides unique advantages. First, in children and adolescents, risk for adverse effects constrains studies of novel therapeutics (11). Adverse effects initiated by ingested substances can be difficult to terminate and can last for hours or days; many effects generated by brain stimulation more rapidly abate, disappearing with cessation of stimulation. Thus, the risk of enduring adverse effects may be lower with brain stimulation than with novel medications. Second, studies of brain stimulation for mood disorders use imaging to leverage cross-species research on neural circuitry function (11). A comparable approach in anxiety could allow imaging studies in youths and experiments in other species to identify neural circuit targets associated with avoidance and other dysfunctional behaviors evoked by threats. These circuits might be targeted through brain stimulation in pediatric patients with refractory anxiety disorders.

Whether through pharmacology, brain stimulation, or other means, long-term treatment advances likely require changes in practice involving closer basic-clinical integration, which minimally informs current guidelines (31, 32). Closer integration is needed because of the degenerate nature of the nervous system (99), which allows multiple circuits to support many specific behaviors. From this perspective, any codification of specific behaviors remains wanting until the code connects these behaviors to particular circuits. In this instance, degeneracy would require dysfunctional behaviors to be classified based on the combination of phenomenology existing alongside specific

neural circuitry correlates. This departs from current nologies based only on phenomenology.

Applied to defensive behavior and anxiety, degeneracy allows approach-avoidance conflicts to evoke biologically distinct forms of avoidance that appear superficially similar on behavioral, phenomenological grounds (7, 65). In these instances, avoidance in some individuals may reflect the brain's intrinsic tendency to overvalue avoidance of threats; in other individuals, avoidance may reflect the brain's intrinsic tendency to undervalue procurement of rewards, and, in a third group of individuals, avoidance may reflect problems in the brain's ability to adapt effectively after prior exposures to threats and rewards. Thus, these three groups of individuals might manifest superficially similar problems with approach-avoidance conflicts that arise from distinct source mechanisms. Such groups could require different treatments, each tailored to particular mechanisms (7, 100). In the long term, such degeneracy requiring distinct mechanism-guided treatments ultimately could create scenarios where appropriate treatment of mental disorders requires that behavior be classified, diagnoses be assigned, and treatments be selected on the basis of assessments of brain function.

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